

SURAJ ROY

Senior Mobile Software Engineer | Flutter Expert | Mobile Architect | Technical Lead

Howrah, India • +91 91237 27623 • spyrogypsertech@gmail.com

Portfolio • LinkedIn • GitHub

PROFESSIONAL SUMMARY

Senior Mobile Software Engineer with 5+ years of experience designing, architecting, and delivering enterprise-grade mobile applications across Android and iOS platforms using Flutter, Dart, native Android (Java, Kotlin, Retrofit), native iOS (Swift), and React Native. Proven expertise in building scalable, secure, and high-performance applications while leading cross-functional engineering teams through the complete software development lifecycle.

Strong background in Clean Architecture, MVVM, MVC, BLoC, Riverpod, GetX, RESTful APIs, native Bluetooth and GPS/location integrations, Firebase, Google Cloud Platform, Node.js, Next.js, Vue.js, .NET Core, SQL, Docker, CI/CD automation, and enterprise mobile security. Experienced in mentoring developers, defining engineering standards, performing architecture reviews, and driving technical excellence across product teams.

Successfully delivered production applications in SaaS, Logistics, Healthcare, E-commerce, Education, and Enterprise domains while collaborating closely with product managers, designers, backend engineers, QA teams, and stakeholders. Passionate about scalable software architecture, reusable component libraries, mobile performance optimization, DevOps automation, AI-powered features, and building world-class mobile products.

CORE COMPETENCIES

Mobile Architecture • Flutter Development • Native Android (Java) & Native iOS (Swift) • React Native • Technical Leadership • Team Mentoring • Clean Architecture • MVVM / MVC / BLoC • CI/CD Pipelines • Cloud Integration • Performance Optimization • Mobile Security • Product Engineering • Agile / Scrum • Code Reviews • System Design • Cross-functional Collaboration

TECHNICAL SKILLS

Languages: Dart • Flutter • TypeScript • JavaScript • Kotlin • Java • Swift • Python • SQL • C#

Mobile: Flutter • React Native • Native Android SDK (Java, Kotlin, Retrofit) • Native iOS (Swift, AV Foundation) • Google ML Kit • Bluetooth (native pairing & hardware integration) • GPS & Location Tracking / Mobility Solutions • Camera • Push Notifications • Deep Linking • Offline Storage (SQLite / Hive / SharedPreferences) • Platform Channels • Responsive UI • Animations

Architecture & State: Clean Architecture • MVVM • MVC • BLoC • Riverpod • GetX • Provider • Repository Pattern • Dependency Injection • SOLID • Design Patterns • Modular Architecture

Backend & Data: Node.js • Next.js • .NET Core (Web API, Entity Framework Core, Dependency Injection, Middleware, Identity/JWT Auth) • React • Vue.js • Firebase • REST APIs • PostgreSQL • MySQL • SQL Server • SQLite • Authentication (JWT / OAuth2)

Cloud & DevOps: Google Cloud Platform (Cloud Functions, Cloud Run, BigQuery) • Firebase • Docker • CI/CD • GitHub Actions • Fastlane • Jenkins • Git

Testing, Analytics & AI: Unit / Widget / Integration Testing • Mockito • Postman • Google Analytics (GA4) • Firebase Analytics • Crashlytics • OpenAI • Google ML Kit • Barcode / OCR

Tools: Jira • Slack • Confluence • Git • Postman • Android Studio • VS Code • Xcode

PROFESSIONAL EXPERIENCE

Assistant Manager – Mobile Engineering | Inland World Pvt Ltd

Kolkata, India

March 2024 – Present

- Lead a team of 5–6 engineers delivering enterprise mobile and web applications using Flutter, native Android (Java, Retrofit, Android SDK), native iOS, and React Native, driving digital transformation initiatives across logistics and field-operations workflows.
- Architected and delivered a suite of industrial mobile applications — an Inland Operations app, a Sales app, and a Driver app — supporting end-to-end docket token generation, validation, and tracking, along with real-time GPS-based mobility and fleet-tracking solutions for field and driver operations.

- Built native Android modules in Java using Retrofit for REST API communication and background sync, and integrated native Bluetooth components for hardware-connected workflows such as docket-token printing and device pairing within the Driver app.
- Define mobile architecture standards using Clean Architecture and MVVM principles across Flutter, native Android (Java), and native iOS codebases to improve scalability, maintainability, and code quality.
- Own the full software development lifecycle across the mobile and web stack — Flutter, React Native, native Android/iOS, Node.js, Next.js, and Vue.js — covering requirements analysis, architecture design, development, testing, deployment, and post-release support using Agile/Scrum practices with Jira and Slack.
- Design and maintain backend services using .NET Core (Web API, Entity Framework Core, Dependency Injection, Middleware) with SQL Server as the primary data store, alongside Node.js and Next.js services powering internal tooling and web-facing applications.
- Establish coding standards, reusable component libraries, and engineering best practices across Java/Android, Swift/iOS, Flutter, React Native, Vue.js, and .NET Core teams; mentor developers through structured code reviews and technical guidance.
- Implement Docker-based environments and CI/CD pipelines (GitHub Actions, Fastlane) to streamline build, test, and release processes across mobile and web deliverables, improving deployment reliability and release cadence.
- Collaborate with product, design, QA, and backend teams to deliver secure, high-performance applications with robust API integrations, JWT/OAuth2 authentication, offline-first capabilities using SQLite, and GPS-based location tracking for field operations.
- Drive mobile product development as Assistant Manager — partnering with stakeholders to define product roadmaps, prioritize feature backlogs, scope releases, and translate business requirements into technical specifications for the Inland Operations, Sales, and Driver apps.
- Own QA strategy and release governance across the mobile portfolio — defining test plans, driving unit/widget/integration test coverage, coordinating UAT with business stakeholders, and gatekeeping release sign-off to ensure production quality.
- Champion automation across the SDLC — building automated build, test, and deployment pipelines (GitHub Actions, Fastlane, Docker) and introducing automated regression and QA test suites to reduce manual testing effort and accelerate release cycles.
- Track team velocity, sprint health, and delivery metrics as part of people and product management responsibilities, providing status reporting to leadership and aligning engineering output with organizational priorities.

Senior Mobile Engineer | Ionio

Delaware, USA (Remote)

November 2023 – February 2024

- Architected and led development of a production SaaS platform for Amazon sellers, enabling advanced product research and retail arbitrage through high-performance barcode scanning.
- Designed cross-platform scanning capabilities by integrating Google ML Kit (Android) and AV Foundation (iOS) within a Flutter application, delivering near-native performance and reliability.
- Built full-stack features using Flutter (frontend), Node.js (backend services), and PostgreSQL; established CI/CD pipelines for automated Android cloud deployments and continuous delivery.
- Owned technical decision-making for mobile architecture, performance optimization, third-party ML integrations, and production readiness standards for e-commerce workflows.

Software Development Engineer | Cloudcraftz Solutions

Kolkata, India

May 2022 – November 2023

- Led end-to-end development of multiple production cross-platform mobile applications (including OptiqueSutra School on Google Play) using Flutter and Firebase, incorporating push notifications, analytics, and reporting.
- Architected backend services with Java (Micronaut), TypeScript Cloud Functions, Python (Flask), and GCP (BigQuery, Cloud Run) to power real-time features, data pipelines, and scalable analytics.
- Applied Clean Architecture principles, GetX state management, MVC patterns, and rigorous API testing (Postman, Mockito) to maintain high-quality, maintainable, and testable codebases.
- Integrated OpenAI endpoints for model generation and fine-tuning; delivered custom analytical dashboards using Google Analytics 4 (GA4) and Firebase Analytics for diverse stakeholder needs.
- Collaborated with design and product teams to ship intuitive, user-centric applications while providing technical mentorship and resolving production issues.

Software Engineering Intern | Craftveda Technology

Kolkata, India

January 2022 – March 2022

- Developed a MedTech mobile application with real-time Bluetooth integration for capturing and analyzing vital health metrics, enabling continuous patient monitoring.
- Built cross-platform functionality with React Native and interactive data visualization using Highcharts JS; successfully published the application to the Google Play Store.

Flutter Developer | Distarise

Mysore, India

January 2021 – July 2021

- Collaborated with senior engineers on production Flutter applications, refining skills in writing efficient, high-quality, maintainable Dart code and applying established mobile development best practices.

SELECTED PROJECTS

Inland Operations, Sales & Driver Apps (Flutter, Native Android/Java, Retrofit, Bluetooth, GPS) — Suite of industrial logistics applications covering docket token workflows, field sales, and driver mobility with real-time GPS tracking.

OptiqueSutra School (Flutter, Firebase) — Education platform published on Google Play Store with push notifications, analytics, and reporting.

Careplix Vitals (React Native, Bluetooth, Highcharts) — MedTech application for real-time vital health metrics analysis, live on Play Store.

Amazon Seller SaaS Platform (Flutter, Node.js, PostgreSQL, Google ML Kit, AV Foundation) — Barcode-scanning product research tool for retail arbitrage.

Additional published apps — X-lent Tea (E-commerce), Sonics Music Station (Streaming), Afterbell (Child progress tracking).

EDUCATION

MBA, Business Analytics | MCKV Institute of Engineering, Howrah

B.Tech, Computer Science Engineering | Techno India University, Kolkata • CGPA: 7.87

12th (ISC) – Sunrise School, CISCE • 73% | 10th (ICSE) – Don Bosco School Liluah, CISCE • 84.8%

ACHIEVEMENTS & RECOGNITION

- Multiple awards in prestigious hackathons including the Google Cloud Solution Challenge.
- 3rd Prize – Maker Mela, recognized for innovative technical solutions.
- Recognized for achievements in RestartIndia, demonstrating strong problem-solving and innovation capabilities.
- Published multiple production applications on Google Play Store spanning e-commerce, education, music streaming, and MedTech domains.